

Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: Non-Arab Tourists and Expats in Eight Arab Countries

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU's 57-subject taxonomy is anchored in US academic curricula with no dedicated Arab history, Middle Eastern geography, Arabic language, or Islamic jurisprudence subjects. Dataset analysis confirms that even subjects nominally relevant to the deployment (high_school_world_history, high_school_geography, global_facts, world_religions) contain little to no Arab-region content in sampled examples. Given IO is rated HIGH priority and the deployment's core domain is precisely what MMLU omits, this is a fundamental construct underrepresentation.

Strengths

- Universal STEM and formal reasoning subjects (mathematics, physics, chemistry, formal logic, computer science) test culturally neutral content valid for any region, including Arab contexts (Strength 1; D1–D43 passim)
- World religions subject nominally covers multiple faith traditions, providing at least a structural slot where Islamic content could appear [Q171]
- History subject claims to span 'a wide range of time periods and geographical locations' [Q41], even if dataset evidence shows minimal Arab-region instantiation

ID	Evidence
D1	The right of American citizens to petition the government for a redress of grievances is protected under the First Amendment — US constitutional law — exclusively US institutional framing
D2	In the majority of cases, federal programs are implemented by state and local governments, by means of federal funding — US federal governance structure
D3	Both the War Powers Act of 1974 and the Budget and Impoundment Control Act of 1974 represent efforts by Congress to limit the powers of the president — US-specific legislative history
D4	John C. Calhoun, 'South Carolina Exposition and Protest,' 1828 — US political history — no Arab-region analog
D5	Tonight, the daughter of an immigrant from Italy has been chosen to run for (vice) president... Geraldine Ferraro, Vice Presidential Nomination Acceptance Address, July 19, 1984 — US domestic politics 1984
D6	A truck driver was assigned to drive a huge tractor-trailer loaded with logs... the girl was struck by the tractor-trailer and was seriously injured — US tort law scenario
D7	A patient filed a medical malpractice action against a hospital in federal court — US federal court procedure
D8	A buyer filed a lawsuit against a seller based on a written contract allegedly executed at the time of the sale of the seller's hot dog stand — US contract law
D9	Why is NSC 68 seen as a turning point in US Cold War Foreign policy? ... It indicated a shift towards military containment — US Cold War policy — framed entirely from US perspective
D10	In American government, the power to declare war rests with Congress. — US constitutional structure
D11	Revelations that the NSA was monitoring the communications of American citizens without obtaining warrants — US domestic surveillance scandal
D12	For which of these two scenarios does the main character do something clearly morally wrong, according to ordinary moral standards in the US as of 2020? — Explicitly frames moral judgment via US 2020 standards
D13	according to ordinary moral standards in the US as of 2020? Scenario 1: When I went through the road block I gave the policeman my brothers ID. — US-anchored moral framing

D14	according to ordinary moral standards in the US as of 2020? Scenario 2: I was due to start work soon, so I went to a baseball game. — US cultural reference (baseball) embedded in moral reasoning
D15	British journalist William Howard Russell, My Indian Mutiny Diary, 1860... 'the deed was done by a subject race — by black men who dared to shed the blood of their masters' — Colonial-era British framing of Indian revolt
D16	Excerpt from the inscription of Kerdīr at Naqsh-e-Rustam, Persia, late third century C.E. ... Jews, Buddhists, Hindus, Nasoreans, Christians, Maktak, and Manichaeans in the empire were smitten — Near Eastern (Persian) source — the only example touching the broader Middle East region
D17	Rome claimed that their bishop (pope) was the direct successor of which leader? ... Peter — Christian ecclesiastical history — no Islamic content in sampled examples
D18	In Jainism, what is the cycle that one must liberate oneself from? ... Samsara — Jainism content
D19	Which are the synoptic Gospels? ... Matthew, Mark, Luke — Christian Biblical content
D20	Which Jaina group only allows women to lead a life of semi-renunciation? ... Digambara — Jainism content — no Islamic content in sample
D21	What is the most important prayer in Judaism? ... The Shema — Jewish religious content
D22	About what percentage of survey respondents from China report having paid a bribe in the last year to access public services ... as of 2017? — China-focused global facts
D23	As of 2017, what fraction of the population in India used the internet in the past three months? ... 26% — India-focused global facts
D24	As of 2019, which of the following had the lowest life expectancy? ... Mexico — Global health — Latin American and Asian countries featured; no Arab countries
D25	As of 2018, about what percentage of Indians say that having an increasing number of people of many different races ... makes it a better place to live? — India-focused survey data
D26	At the time of independence, there were already hundreds of thousands of university graduates in ____, but hardly any at all in _____. ... India, Congo — Development economics — Africa/Asia framing, no Arab countries
D27	What is the most rapidly growing religion in the United States today? ... Islam — US-context geography question; Islam mentioned but in US demographic context
D28	Which zone contains low-income slums, ethnic ghettos, and general deterioration in Burgess's concentric zone model of urban form? — US urban sociology model (Burgess)
D29	Leeds Woolen Workers Petition, 1786 ... The sentiments expressed by the Leeds woolen workers illustrate which of the following historical trends? — British industrial history
D30	Voltaire, Letters on the English Nation, 1733 ... of all religions, the Christian ought doubtless to inspire the most tolerance — European Enlightenment philosophy
D31	civil society is far less developed than in, for instance, _____ ... Russia, China, Britain — Western-centric civil society framing; Arab world absent from options
D32	The age at which one can receive full benefits from social security ... Has been increasing — US social security system
D33	Which of the following groups of Americans have the lowest rate of alcohol abuse? ... Asian-American — US demographic framing
D34	Which of the following is the most common method of birth control for married couples in the U.S. ... Sterilization — US-specific reproductive health statistics
D35	From 1988 to 1990 among heterosexuals in the US, the number of unmarried adults aged 20 to 45 who report having multiple partners has: declined for both men and women — US sexual behavior survey data
D36	Archaeological evidence indicates that cattle were first domesticated where and how long ago? ... in the Middle East, about 10,500 years ago — Middle East referenced as origin of cattle domestication — one of very few regional mentions
D37	Kwame Nkrumah, Neo-Colonialism, 1965 ... 'A State in the grip of neo-colonialism is not master of its own destiny.' — African post-colonial context — non-Western source, though not Arab
D38	Which of the following statements best describes the postmodernist project? ... It rejects the Kantian concern with individual rights, equality, and justice. — Western jurisprudential theory; no Islamic jurisprudence
D39	What is the meaning of cultural relativism? ... Cultural relativism posits that local culture should validate the existence and practice of all human rights — International law — potentially relevant to Arab regional norms
D40	Jim watched a liquor store furtively for some time, planning to hold it up ... On a charge of burglary, Jim's best defense would be that the liquor store was open to the public. — US criminal law scenario
D41	As reported on CNN, in a May 1999 national poll 43% of high school students expressed fear about going to school — US-sourced statistics example

D42	Best described as intensification of worldwide social relations and increasing interdependence, globalization is the result of the compression of space and time — Abstract IR theory — not regionally specific
D43	With regard to minority and nonminority clients, psychotherapy is ... equally effective — US psychology research context
Q171	“Figure 70: A World Religions example.” (p.27)
Q41	“Finally, our history questions cover a wide range of time periods and geographical locations, including prehistory and other advanced subjects.” (p.4)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories per the deployment YAML and elicitation include Arab history (Islamic Golden Age, Ottoman period, modern state formation), regional geography of eight countries, Arabic language basics, Islamic religious knowledge, and pan-Arab cultural background.

IO-2: Yes — MMLU has no dedicated subject for any of these categories. The 57 subjects include `high_school_us_history`, `high_school_european_history`, `us_foreign_policy`, `professional_law` (US bar exam), but no Arab/MENA equivalents [Q1, Q4, Q37, Q43]; dataset confirms exhaustively (D1, D3, D4, D9–D11).

IO-3: Yes — multiple subjects (`high_school_us_history`, `high_school_government_and_politics`, `us_foreign_policy`, `professional_law`, `human_aging`, `human_sexuality`) are US-institution-specific and contribute construct-irrelevant variance for an Arab-region deployment (D1–D11, D32–D35).

IO-4: Confirmed structural gap: zero dedicated Arab-region subjects out of 57; nominally relevant subjects (`global_facts`, `world_religions`, `high_school_geography`) contain little or no Arab-region content in 5–6 sampled examples each (D22–D28). ArabicMMLU and ILMAAM exist as supplements but are not part of MMLU [WEB-2, WEB-5].

ID	Evidence
WEB-2	ArabicMMLU is the closest available substitute for the Arabic-language secondary cohort (14,575 MSA questions) (https://aclanthology.org/2024.findings-acl.334/)
WEB-5	ILMAAM (2025) identified significant cultural misalignments in religion and morality content; 11 expert annotators reviewed for cultural appropriateness, bias, religious sensitivity (https://aclanthology.org/2025.loreslm-1.29/)
Q1	“We propose a new test to measure a text model’s multitask accuracy. The test covers 57 tasks including elementary mathematics, US history, computer science, law, and more.” (p.1)
Q37	“Branches of the humanities include law, philosophy, history, and so on (Appendix B).” (p.4)
Q43	“Subject areas include economics, sociology, politics, geography, psychology, and so on.” (p.4)
Q169	“Figure 68: A US Foreign Policy example.” (p.26)
Q4	“The benchmark covers 57 subjects across STEM, the humanities, the social sciences, and more. It ranges in difficulty from an elementary level to an advanced professional level, and it tests both world knowledge and problem solving ability.” (p.1)
D9	Why is NSC 68 seen as a turning point in US Cold War Foreign policy? ... It indicated a shift towards military containment — US Cold War policy — framed entirely from US perspective
D10	In American government, the power to declare war rests with Congress. — US constitutional structure
D11	Revelations that the NSA was monitoring the communications of American citizens without obtaining warrants — US domestic surveillance scandal

D1	The right of American citizens to petition the government for a redress of grievances is protected under the First Amendment — US constitutional law — exclusively US institutional framing
D3	Both the War Powers Act of 1974 and the Budget and Impoundment Control Act of 1974 represent efforts by Congress to limit the powers of the president — US-specific legislative history
D4	John C. Calhoun, 'South Carolina Exposition and Protest,' 1828 — US political history — no Arab-region analog
D2	In the majority of cases, federal programs are implemented by state and local governments, by means of federal funding — US federal governance structure
D5	Tonight, the daughter of an immigrant from Italy has been chosen to run for (vice) president... Geraldine Ferraro, Vice Presidential Nomination Acceptance Address, July 19, 1984 — US domestic politics 1984
D6	A truck driver was assigned to drive a huge tractor-trailer loaded with logs... the girl was struck by the tractor-trailer and was seriously injured — US tort law scenario
D7	A patient filed a medical malpractice action against a hospital in federal court — US federal court procedure
D8	A buyer filed a lawsuit against a seller based on a written contract allegedly executed at the time of the sale of the seller's hot dog stand — US contract law
D32	The age at which one can receive full benefits from social security ... Has been increasing — US social security system
D33	Which of the following groups of Americans have the lowest rate of alcohol abuse? ... Asian-American — US demographic framing
D34	Which of the following is the most common method of birth control for married couples in the U.S. ... Sterilization — US-specific reproductive health statistics
D35	From 1988 to 1990 among heterosexuals in the US, the number of unmarried adults aged 20 to 45 who report having multiple partners has: declined for both men and women — US sexual behavior survey data
WEB-12	Survey of Arabic benchmarks notes gaps in coverage of North Africa, Levant, Iraq even within the Arabic NLP ecosystem (https://arxiv.org/html/2510.13430v1)
D22	About what percentage of survey respondents from China report having paid a bribe in the last year to access public services ... as of 2017? — China-focused global facts
D23	As of 2017, what fraction of the population in India used the internet in the past three months? ... 26% — India-focused global facts
D24	As of 2019, which of the following had the lowest life expectancy? ... Mexico — Global health — Latin American and Asian countries featured; no Arab countries
D25	As of 2018, about what percentage of Indians say that having an increasing number of people of many different races ... makes it a better place to live? — India-focused survey data
D26	At the time of independence, there were already hundreds of thousands of university graduates in ____, but hardly any at all in _____. ... India, Congo — Development economics — Africa/Asia framing, no Arab countries
D27	What is the most rapidly growing religion in the United States today? ... Islam — US-context geography question; Islam mentioned but in US demographic context
D28	Which zone contains low-income slums, ethnic ghettos, and general deterioration in Burgess's concentric zone model of urban form? — US urban sociology model (Burgess)

Information Gaps

- No published quantitative audit of original English MMLU for Arab-region content proportion across all 15,908 items (FV1)
- Whether the full world_religions set (171 questions) contains Islamic content beyond the sample

Requires Expert Verification

- Full-set count of Arab-region or Islamic-content questions across world_history, world_religions, global_facts, and miscellaneous subjects

Remediation

Gap 1: No dedicated Arab history, Middle Eastern geography, Arabic language, or Islamic content subjects in MMLU's 57-subject taxonomy.

Recommendation: Replace MMLU as the primary Arab-region knowledge evaluation with ArabicMMLU [WEB-2] (40 tasks across 8 Arab countries) and PalmX 2025 [WEB-13] (Arabic and Islamic culture). Retain MMLU only for transferable

STEM/formal-reasoning subjects.

ID	Evidence
WEB-2	ArabicMMLU is the closest available substitute for the Arabic-language secondary cohort (14,575 MSA questions) (https://aclanthology.org/2024.findings-acl.334/)
WEB-13	https://arxiv.org/html/2509.02550